Enedril

Fishing and Drilling Solutions

DIRECTIONAL DRILLING & FISHING SOLUTIONS

FRANCE

ABOUT US

Enedril (formerly Mustang Services) is an independent French company founded in 2013 and specialised in directional drilling, fishing and well intervention for geothermal, oil & gas and underground gas storage projects, in France and abroad. We combine solid field experience with technical know-how, and we are known for being fast, flexible and easy to work with.

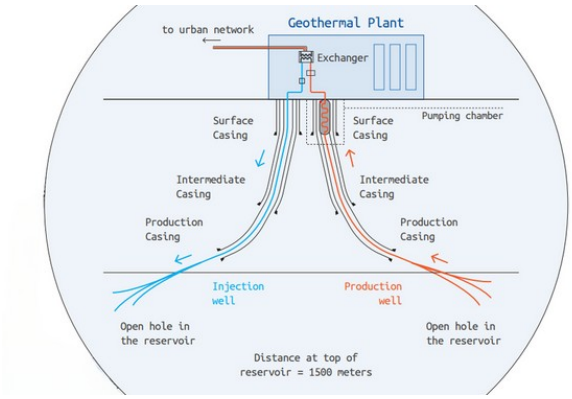
98,500 m	60	142	2,500
drilled	wells, up to 3,000 m TVD	fishing, workover and P&A jobs	tools in stock

OUR FACILITIES

Since 2021 we operate from Lons, near Pau, with two fully equipped workshops of more than 1,000 m², a 5 t overhead crane, a break-out unit, a jar tester, lathe, milling and grinding machines and a separate welding shop.

A 3,500 m² outdoor yard stores tubulars, stabilizers and on-site containers, for large-scale logistics.

A rare equipment stock in France: mud motors from 3-1/8" to 9-5/8", shock tools, drilling jars, MWD kits and a very wide range of fishing tools.



Geothermal doublet: injection and production wells

OUR VALUES

RESPECT
For our clients, our teams and the people and places around us.

TEAMWORK
Real success comes from strong values and working together.

ENVIRONMENT
Caring for the environment is at the core of what we do.

ISO 9001:2015 CERTIFIED

A quality management system that follows every job from start to finish: safety, quality and reliability on each intervention.

About 15 clients supported each year, in France and internationally.

WHY CHOOSE US

EXPERIENCE
Years in the industry and a wide range of projects completed across sectors.

EXCELLENCE
Dedication to excellence, safety and innovation for superior results.

CUSTOM SOLUTIONS
Every project is unique: tailored tools and procedures.

RELIABILITY
A partner ready to go the extra mile to meet your needs and deadlines.

OUR PURPOSE

Supporting the next generations in the energy transition by contributing to the production of more responsible energy.

DIRECTIONAL DRILLING & MWD

Enedril provides advanced directional drilling solutions for complex wells with high precision requirements: downhole motors (PDM), rotary steerable systems (RSS) and high-temperature MWD tools. Experienced Directional Drillers and MWD Engineers steer and optimise the trajectory continuously, with well planning done in-house.

3-1/8" to 9-5/8"
PDM motors, high-temperature power sections

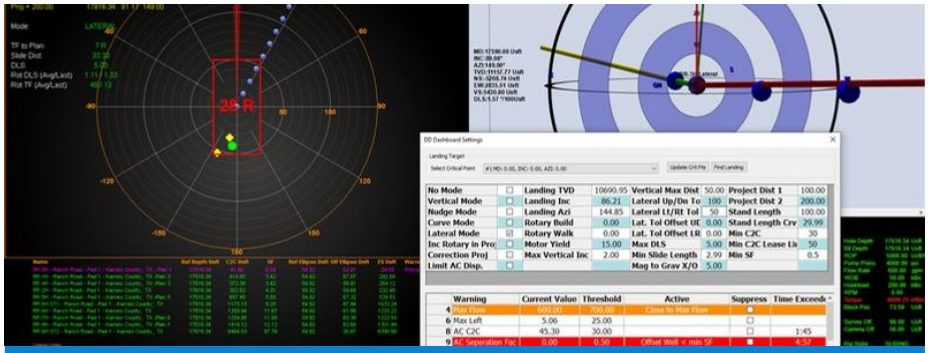
RSS
Push-the-bit and point-the-bit systems

up to 200°C
MWD kits with HT electronics



MUD MOTORS & RSS

PDMs from 3-1/8" to 9-5/8" OD with a wide range of power sections, including high-temperature stators. Push-the-bit and point-the-bit RSS for precise control in complex formations.



WELL PLANNING

Innova and DrillScan software: torque and drag, hydraulics and anti-collision, to drill safely and efficiently in line with project goals. Full simulation, high-frequency data recorder.

BOOST MUD PULSE MWD SYSTEM

Rock solid performance for aggressive drilling conditions. A positive mud pulse system that sends surveys, directional telemetry, gamma ray and other sensor data to surface in real time.



- Field-proven design, standard rating 185°C, 200°C version available
- Top and bottom mounted positive pulse systems
- Motor-driven pulser with anti-plugging technology: performs in LCM and high-solids mud
- Wide range of flow rates and rotary speeds, easy to assemble and service
- Down-linking for adjustable logging sequences (4 programmable)

OUTSIDE DIAMETER **47.6 mm (1.875")**	PRESSURE RATING **20,000 psi**	TEMPERATURE **-20 to 185°C**	COLLAR SIZES **3.5" to 8"**
FLOW RATE **75 - 1,200 gpm**	BATTERY LIFE **400+ h (2 modules)**	SHOCK LIMIT **500 g**	INTERNAL MEMORY **24 MB**

Telemetry: inclination, azimuth, toolface, continuous inclination and azimuth, gamma ray, shock, vibration, RPM, stick-slip, temperature, annular pressure, battery status. Inclination ±0.10°, azimuth ±0.50°, toolface ±0.20°.

ALSO AVAILABLE

- Survey equipment (LWD, gyro, dual telemetry)
- Turbodrilling
- Jar placement
- Non-magnetic equipment
- Slimhole tools 3-1/2"
- High-temperature systems

98,500 m drilled · 60 wells · up to 3,000 m TVD · geothermal, oil & gas, underground gas storage

FISHING, WORKOVER & PLUG AND ABANDONMENT

Fishing, workover and plug & abandonment (P&A): the full range of operations to recover, restore and secure wells, during drilling, workover or abandonment. Our fishing team responds quickly with a large rental and sale fleet, backed by our own workshop and custom tool design.

2,500	24 / 48h	142	ISO 9001
tools available	response time	fishing, workover and P&A interventions	certified quality management

FISHING

Recovery of rods, drill strings, BHAs and stuck or lost tools.

Jarring and impact tools, shock tools and intensifiers.

Overshots, spears, die collars, magnets, taper taps.

Tools on site for rental or purchase.

WORKOVER

Well restoration and preparation for return to production.

Section milling and wash over.

Casing repair, scrapers, impression blocks, junk baskets.

Enlargement and reaming with RTU, DTU and SPX underreamers.

PLUG & ABANDONMENT

Safe, definitive abandonment of wells.

Cutting with hydraulic pipe cutters and section mills.

Whipstocks and sidetracks, packers, bridge plugs, liner hangers.

Cold cutter, wellhead replacement, casing jack, tanks.









WATER WELLS TOO

We also work on water wells, typically between 300 and 1,000 m, in support of drilling contractors. Habits formed on 3,000 m wells mean we handle shallow wells with the same rigor.

CUT & PULL

Hydraulic and external pipe cutters, spears and overshots to cut and pull casing, tubing or stuck strings.

REAMING

Rotary and directional underreamers, hole openers, roller reamers and SPX drag-type underreamers.

MILLING

Junk mills, pilot and taper mills, section mills and washover shoes to clear and open the hole.

WHERE WE WORK

Based in Lons (64), south-west France. We mobilise across metropolitan France within 24 to 48 hours in an emergency, and have international experience in Europe and North Africa. A single dedicated team follows each job from diagnosis to resolution.

TOOL CATALOGUE

A selection of the main tools in stock, available for rental or purchase. The full catalogue and technical data sheets are available on request. Sizes in inches; tool series are quoted where applicable.

CLEANING	Range
Casing scraper	3-5/8" - 7-7/8"
Impression block	2-1/8" - 13-1/2"
Reverse circ. junk basket	5-5/8" - 17"
Junk basket	3-1/2" - 13-3/8"

MILLING & WASHOVER	Range
Economill	3-3/4" - 12-1/8"
Junk mill	2-1/8" - 17-3/8"
Super junk mill	3-3/4" - 18-3/8"
Pilot mill	3-1/4" - 8-1/2"
Taper mill	2-1/2" - 18-1/2"
Washover shoe	2-9/16" - 13-3/8"

REAMING & ENLARGEMENT	Range
Rotary underreamer (RTU)	DTS 350, 5700, 8200
Drilling-type underreamer (DTU)	5700, 8200, 12000, 18000
Master Driller	8200, 14000
Hole opener	14-3/4"
SPX drag-type underreamer	5700
Roller reamer	8200

CUTTING	Range
Hydraulic pipe cutter	3600, 5700, 8200, 11700s
Section mill	3600, 5500, 8200, 11700s
External hydraulic cutter	5-5/8" - 8"

RECOVERY	Range
Die collar	4" - 9-5/8"
Fishing magnet	3-3/4" - 15"
Spear, internal	6-3/4" - 8"
Spear	2-3/8" - 20"
Overshot	SFS, SH, XSH, FS
Taper tap	3-1/16" - 8"

JARRING & IMPACT	Range
Fishing jar	3-1/8" - 8"
Jar intensifier	3-1/8" - 6-1/2"
Safety joint	3-1/2" - 8"
Stabilizer	3-1/8" - 9-1/2"
Crossover (XO)	3-3/8" - 11-3/4"
Bit sub / Float sub / Bull nose	2-3/16" - 10-1/4"

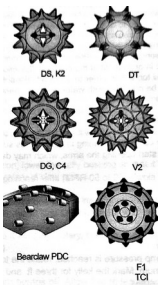
FOR WATER WELLS (typically 300 - 1,000 m)

Cut & pull: hydraulic pipe cutters, external cutters, spears, overshots, jars.

Reaming: RTU, DTU, SPX drag-type, hole openers, roller reamers.

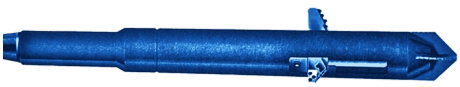
Milling: junk mills, pilot mills, taper mills, section mills, washover shoes.

Stock available, response within 24/48h anywhere in metropolitan France.

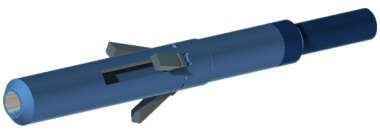


UNDERREAMERS

DTU and SPX data sheets available



- **Downhole tools and bits:** drilling bits (PDC), underreaming, jars, stabilizers, tubulars, circulating subs.
- **Special interventions:** whipstocks and sidetracks, packers, bridge plugs, liner hangers, casing jack, tanks.
- **Custom tools:** designed with an API 5 & 7 certified machine shop, repaired and redressed in our workshop.



HYDRAULIC PIPE CUTTER

Data sheet available

- Internal hydraulic cutter for single or multiple strings of concentric or eccentric casing.
- Mainly used in well abandonment, when casings are cemented and cannot be cut and pulled individually.
- Four tool diameters (3-5/8", 5-3/4", 8-1/4", 11-3/4") cutting casing from 4-1/2" to 36".
- Operates up to 400 L/min and 35 - 60 bar. Knives: standard, hardfacing or carbide.

HOW TO REQUEST A QUOTATION: specify the tool type, size or series, top and bottom connections, and the operation planned (cut & pull, reaming, milling...). Rental or purchase.

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Fishing response within 24/48h across metropolitan France